Experiment 08:

Name of the Experiment: Voltage Divider Biasing of Enhancement Type MOSFET

Theory:

MOSFETs

A Metal–Oxide–Semiconductor Field Effect Transistor (MOSFET) is a type of Field Effect Transistor (FET) that operates with voltage control at the gate terminal. Unlike BJTs, which are current-controlled devices, MOSFETs use an electric field induced by the gate-to-source voltage (V_{GS}) to control the current flowing from drain to source (I_D). The MOSFET has three terminals: Gate (G), Drain (D), and Source (S). In an n-channel Enhancement-type MOSFET (E-MOSFET), no drain current flows when V. Conduction begins only when. Above this threshold, the drain current follows the quadratic relationship:

$$I_D = k(V_{GS} - V_{th})^2$$

Here,
I_D is the drain current.
The V_{GS} is the Gate to Source Voltage.
The V_{th} is the threshold voltage at which the MOSFET turns on
k is the transconductance parameter which solely depends on the type MOSFET being used.

Biasing Techniques

To set a proper operating point (Q-point) for MOSFET circuits, several biasing techniques can be used.

The Self Biased Arrangement (Feedback biasing arrangement): This circuit, designed in figure 7.1, shows how the self-biased circuit works. The idea is to connect a very high resistance between the drain and gate which can draw huge of voltage and supply it to the gate terminal. Since the current $I_G = 0\ A$, the current in flowing through R_D remains the same.

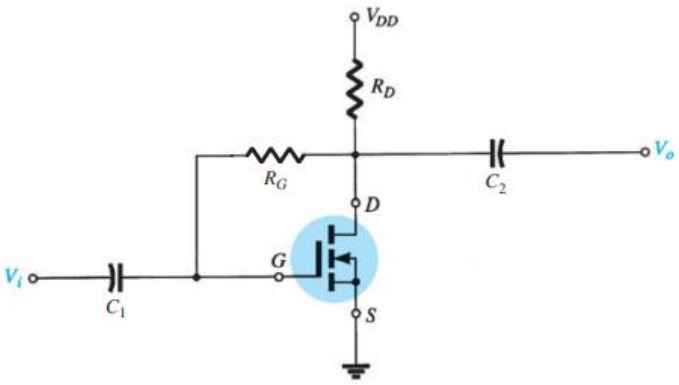


Fig 7.1:
Feedback Biasing Arrangement

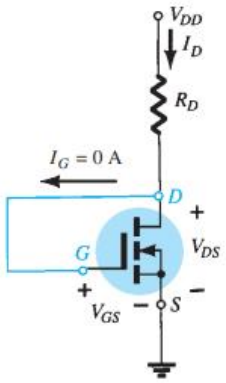


Fig 7.2:
DC equivalent of Fig 7.1

Voltage Divider Biasing Circuit: The circuit given in Figure 7.3, shows how the voltage divider circuit works. What this circuit does is that since $I_G = 0\ A$, it takes the voltage V_{DD} and divides it amongst R_1 and R_2, thus, using Voltage Divider Rule, the gate voltage (V_G) applied will be equal to the voltage across R_2.

$$V_G = \frac{R_2}{R_1 + R_2} V_{DD}$$

Using KVL in the given loop in Figure 7.3, we can get the following equation:

$$V_{GS} = V_G - I_D R_S$$

Recall that drain current (I_D) flowing through R_D and R_S are the same.

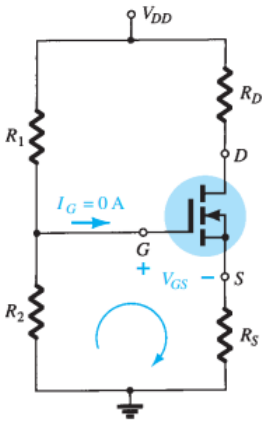


Fig 7.3:
Voltage Divider Biasing Arrangement

Transfer Curve

The Transfer Curve for an n-type MOSFET can be found using Shockley's equation, the graph in Figure 7.4 shows a sudden exponential increase of drain current when the gate-to-source voltage applied is greater than the threshold voltage ($V_{GS} > V_{th}$).

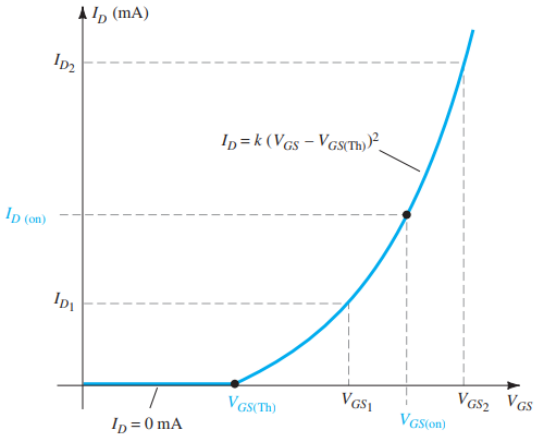


Fig 7.4:
Transfer Curve for n-channel
Enhancement type MOSFET

This curve is obtained after knowing the parameter k, which can be found using the equation:

$$k = \frac{I_{D(on)}}{\left(V_{GS(on)} - V_{th}\right)^2}$$

The drain current and gate voltages needed to calculate k, are taken from the data sheet of the MOSFET being used.

Q-Point

The Q-point (short for Quiescent Point) is the operating point of a transistor when no input signal is applied. For this lab, it represents the steady-state values of drain current (I_D) and gate-to-source voltage (V_{GS}) that are set by the biasing network.

Which means, after building the biasing circuit in the lab, your measurements will be your Q-point values. However, it is also important to find the theoretical Q-point of a MOSFET from the transfer curve by applying load line analysis as shown in Figure 7.5.

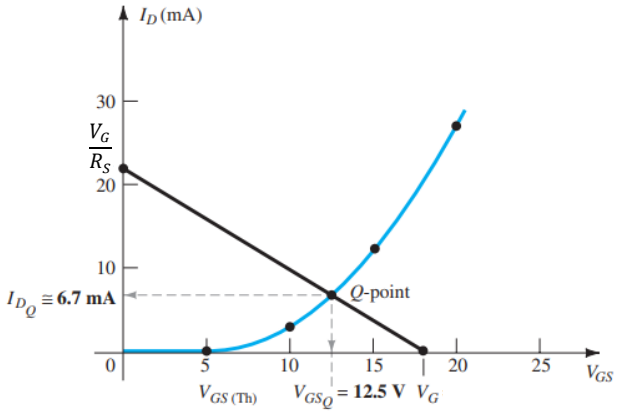
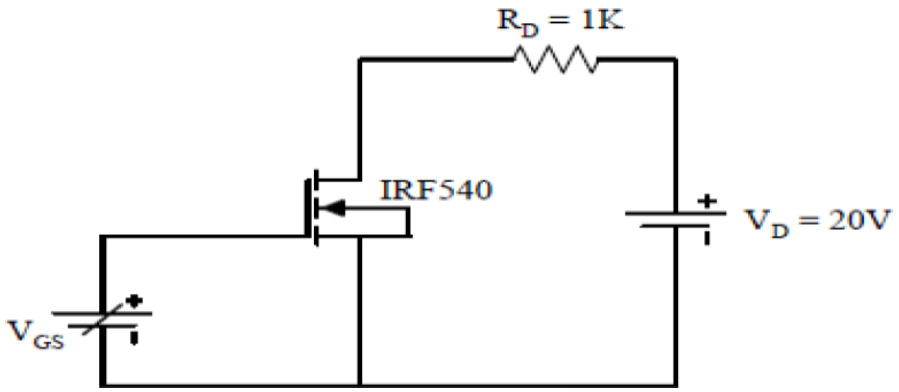


Fig 7.5:
Load Line Analysis on the Transfer Curve for n-channel Enhancement type MOSFET

Equipment and Component:

Serial no.	Component Details	Specifications	Quantity
1	MOSFET	IRF540	1 piece
2	Resistor	1kΩ, 470Ω, 5kΩ	1 piece each
3	Potentiometer (POT)	10kΩ	1 piece
4	Trainer Board		1 unit
5	DC Power Supply		1 unit
6	Digital Multimeter		1 unit
7	Chords and Wires		1 unit

Experimental Setup 1:



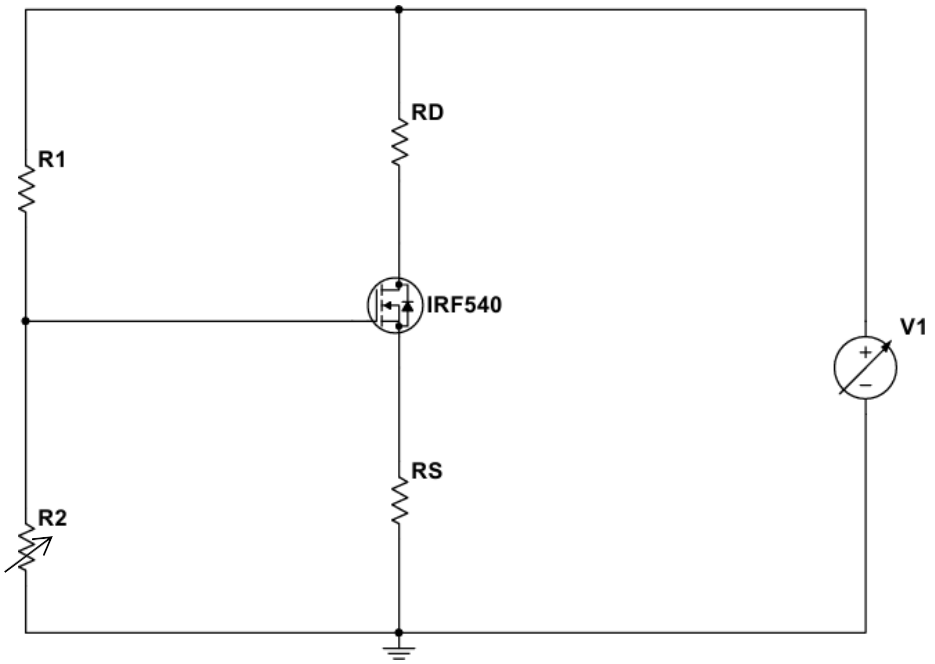
Procedure:

1. Build the circuit as per the setup.
2. Set V_{GS} to zero and $V_{DS} = 15V$ measure I_D.
3. Increase the gate voltage V_{GS} gradually and record the readings.
4. Take reading until I_D suddenly increases and goes to 15 mA.
5. Note the value of V_{GS}, and I_D (approximately 15 mA) and write it down in Table 7.1.
6. Calculate the value of k.

Table 7.1: Calculation to find k

$V_{GS(on)}$	$I_{D(on)}$	V_{th}	$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{th})^2}$
	10 mA	3 V	

Experimental Setup 2:



Procedure:

1. Connect Resistors R1 = 5kΩ, RD=1kΩ, RS=470Ω and set up the circuit as shown.
2. For R2, connect a 10kΩ potentiometer and set it to minimum, and set V_1=18V.
3. Measure V_G by putting one probe of the DMM to Gate terminal and the other probe to ground.
4. Gradually Increase R2 until VG = 8V.
5. Note the value of V_{GS}, and I_D and write it down in the table.

Table 7.2: Finding the Q-points

Measured		Calculated	
V_{GSq}	I_{Dq}	V_{GSq}	I_{Dq}

Report:

1. Draw the transfer curve based on the data that you have calculated in Table 1.
2. Find the Q-points from your graph and check whether your Measured data matches with the data you found.
3. Does the Shockley equation depend on the type of biasing circuit used or the type of MOSFET being used? Explain.